

Bell Inequality Violations from Non-Injective Projection

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Abstract

Violations of Bell inequalities are commonly interpreted as evidence for either nonlocal dynamics or the failure of classical realism. In this work, we show that Bell-type correlations can arise without invoking superluminal influences, retrocausality, or local hidden variables, provided that the mapping between underlying configurations and observable outcomes is non-injective.

We study a class of effective models in which multiple underlying states correspond to the same observable description. Such non-injective projections generically induce non-factorizable joint probability structures at the level of observables, even when the underlying description itself is fully local and deterministic. As a result, the statistical assumptions required to derive Bell inequalities are violated at the effective level, while locality is preserved at the underlying level.

Within this framework, quantum correlations emerge as structural features of projected descriptions rather than as signatures of nonlocal causation. We clarify the precise assumption of Bell's theorem that fails in this setting and show how classical behavior is recovered in regimes where the effective projection becomes approximately injective, suppressing nonlocal correlations through decoherence and environmental coarse-graining.

Our results suggest that Bell inequality violations need not reflect a fundamental nonlocality of nature, but may instead indicate intrinsic information loss induced by non-injective mappings between underlying configurations and observable descriptions.

Keywords: Bell inequalities, quantum correlations, non-injective projection, probabilistic factorization, locality, quantum foundations

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1 Introduction

The experimental violation of Bell inequalities stands as one of the most robust and conceptually challenging features of quantum theory. Repeated tests have confirmed that correlations observed between spacelike separated measurement outcomes cannot be reproduced by any local hidden-variable model satisfying statistical factorization. This empirical fact is commonly interpreted as requiring either nonlocal dynamics, the abandonment of realism, or more radical departures from classical causal intuition.

Despite these conclusions, it is important to recall that Bell inequalities do not constrain physical dynamics directly. They constrain the logical and probabilistic structure of effective descriptions, specifically the assumption that joint outcome probabilities factorize into independent local contributions conditioned on a complete specification of relevant variables. The failure of Bell inequalities therefore indicates a breakdown of effective factorization, but does not, by itself, mandate superluminal influences or causal nonlocality.

In this work, we explore a general and minimal mechanism by which Bell-type correlations may arise without invoking nonlocal dynamics, retrocausality, or hidden variables. The central observation is that effective physical descriptions often arise through many-to-one mappings from underlying configurations to observable outcomes. When this mapping is non-injective, distinct underlying configurations correspond to

the same observable description, leading generically to a loss of factorization at the level of observable probabilities.

The non-injective mapping considered here should not be understood as a purely abstract probabilistic construction. Rather, it reflects a structural projection from an underlying relational configuration space to an effective four-dimensional spacetime description in which observations are defined. Observable events correspond not to individual microscopic configurations, but to equivalence classes of underlying states that cannot be distinguished within the effective description.

We study a class of effective models in which observable outcomes are obtained through such non-injective projections from an underlying configuration space. The underlying description is assumed to be local and deterministic, but is not fully recoverable from observables due to the many-to-one nature of the projection. As a result, joint probabilities of outcomes do not admit a factorized representation, even though no superluminal interaction or exchange of information occurs. The violation of Bell inequalities is thus traced to a structural property of the descriptive mapping itself, rather than to any dynamical nonlocality.

The effective loss of information associated with this projection is not arbitrary. In physical realizations, it is naturally controlled by Planck's constant $\hbar$, which sets the minimal resolution at which distinct underlying configurations can be operationally distinguished. Quantum states therefore correspond to equivalence classes whose typical size is determined by $\hbar$, rather than to sharply defined microscopic configurations. The breakdown of factorization reflects this irreducible coarse-graining, not an epistemic lack of knowledge.

Within this framework, quantum correlations are understood as manifestations of global descriptive consistency. Spacelike separated measurements correspond to different local realizations compatible with a single non-factorizable effective description, rather than to independent subsystems endowed with pre-existing local properties. The assumption of outcome determinism conditioned on local variables fails, not because of hidden influences, but because no complete set of outcome-defining variables exists at the effective level.

This perspective naturally accounts for the emergence of classical behavior. In regimes where environmental interactions, coarse-graining, or loss of resolution render the effective mapping approximately injective, observable descriptions become effectively factorizable and Bell violations are suppressed. Classical statistical independence thus appears as a limiting case of effective injectivity, rather than as a fundamental principle.

The scope of the present work is deliberately restricted. We do not propose a modification of quantum dynamics, introduce additional physical degrees of freedom, or advance a specific underlying dynamical model. Our analysis is structural in nature and concerns the logical conditions under which Bell-type inequalities apply to effective descriptions. By isolating non-injective projection as a sufficient mechanism for the failure of factorization, we aim to clarify the conceptual origin of quantum correlations while preserving both operational locality and the empirical content of quantum theory.

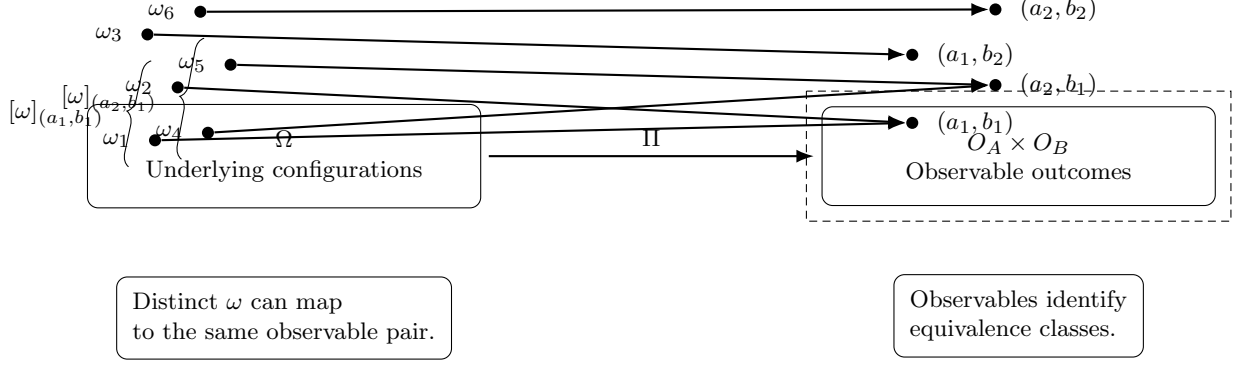


Fig. 1 Schematic non-injective projection from the underlying configuration space Ω to the observable outcome space $\mathcal{O}_A \times \mathcal{O}_B$. Multiple underlying configurations correspond to the same observable pair (a, b) , defining equivalence classes. This many-to-one identification induces a generic failure of probabilistic factorization at the observable level without requiring nonlocal dynamics.

2 Non-Injective Projection and the Failure of Bell Factorization

We consider a minimal abstract setting in which observable descriptions arise from an underlying configuration space through a generally many-to-one mapping. Let Ω denote the space of underlying configurations, assumed to be fully specified and governed by local dynamics. Observable outcomes are defined through a projection

$$\Pi : \Omega \rightarrow \mathcal{O}, \quad (1)$$

where $\mathcal{O}$ is the space of observable states or measurement outcomes. Crucially, Π is assumed to be non-injective, so that multiple distinct elements of Ω correspond to the same observable description.

Figure 1 provides a schematic illustration of this many-to-one projection mechanism. A concrete toy model making this structure explicit is presented in Appendix B.

Let A and B denote two spacelike separated measurement settings, with corresponding observable outcomes $a \in \mathcal{O}_A$ and $b \in \mathcal{O}_B$. In standard local hidden-variable models, one assumes the existence of a variable λ such that the joint probability distribution factorizes as

$$P(a, b \mid A, B, \lambda) = P(a \mid A, \lambda) P(b \mid B, \lambda). \quad (2)$$

Bell inequalities follow from this factorization together with statistical independence assumptions on λ .

In the present framework, no such factorization is generically available. Because the observable outcomes are obtained through the projection Π , the probability distribution accessible at the effective level is given by

$$P(a, b \mid A, B) = \sum_{\omega \in \Pi^{-1}(a, b)} \mu(\omega), \quad (3)$$

where μ is a probability measure on Ω , and $\Pi^{-1}(a, b)$ denotes the set of underlying configurations compatible with the joint observable outcome (a, b) .

When Π is non-injective, the preimage $\Pi^{-1}(a, b)$ does not decompose into independent subsets associated with A and B . As a result, the joint probability $P(a, b | A, B)$ does not admit a representation as a product of marginal probabilities conditioned on any set of effective variables. The failure of factorization arises from the structure of the projection itself, not from any dynamical interaction between the measurement regions.

Importantly, locality at the underlying level is preserved. The measure $\mu(\omega)$ may be defined over configurations with strictly local causal structure, and no superluminal influence or signal exchange is required. The violation of Bell inequalities therefore reflects a mismatch between the factorizable structure assumed in Bell-type derivations and the actual structure of effective descriptions induced by non-injective mappings.

This observation identifies non-injectivity as a sufficient condition for the failure of Bell factorization. Bell inequalities constrain only those effective models in which the mapping from underlying configurations to observables is injective or admits an equivalent factorizable representation. When this condition fails, Bell-type correlations may arise without violating local causality at the underlying level.

3 Bell Inequalities Revisited: Structural vs Dynamical Assumptions

Bell's theorem is commonly presented as a no-go result for local realist theories. More precisely, it establishes that no model satisfying a specific set of probabilistic assumptions can reproduce all quantum correlations. It is therefore essential to distinguish carefully between the assumptions entering the derivation of Bell inequalities and the physical principles they are often taken to represent.

The derivation of Bell inequalities relies on three key assumptions. First, outcome probabilities are assumed to be well-defined and conditioned on a complete specification of relevant variables. Second, joint probabilities are assumed to factorize for spacelike separated measurements, reflecting the absence of direct causal influence between distant measurement settings. Third, the conditioning variables are assumed to be statistically independent of the measurement choices. Violations of Bell inequalities imply that at least one of these assumptions fails.

In many discussions, the failure of Bell inequalities is interpreted as evidence for a breakdown of locality [1]. However, locality in Bell's sense is a condition on the factorization of effective probabilities, not a direct statement about the causal structure of the underlying theory. Bell inequalities constrain models in which observable outcomes can be represented as functions of local settings and a shared set of variables that fully determine those outcomes. They do not directly constrain the existence or absence of superluminal interactions.

Within the framework considered here, the failure of Bell inequalities originates from the breakdown of effective factorization. As shown in the previous section, non-injective mappings between underlying configurations and observable outcomes generically prevent the construction of a complete set of outcome-defining variables at

the effective level. Although the underlying description may be local and deterministic, the observable description does not admit a factorized probabilistic representation.

This distinction clarifies the sense in which Bell's theorem applies. Bell inequalities are valid only for effective descriptions in which observable outcomes can be parameterized by a set of variables that renders the projection from underlying configurations injective or equivalently factorizable. When this condition is not met, the assumptions required for the derivation of Bell inequalities are violated independently of any consideration of causal nonlocality.

From this perspective, violations of Bell inequalities do not compel the introduction of nonlocal dynamics or retrocausal mechanisms. They instead signal a structural limitation of effective descriptions in which observable outcomes are treated as independently specifiable local properties. Nonlocal correlations arise because joint outcomes must be globally consistent with a single non-factorizable effective description, not because information propagates between distant regions.

The present analysis therefore reframes Bell's theorem as a constraint on a class of descriptive models rather than as a direct statement about the ontology or dynamics of physical reality. Bell inequalities test the compatibility of observed correlations with injective and factorizable effective descriptions. When non-injective projections are involved, Bell-type correlations may arise while preserving locality at the underlying level.

4 Measurement, Decoherence, and the Emergence of Classicality

The structural origin of Bell-type correlations identified above naturally raises the question of their apparent absence in classical macroscopic phenomena. Everyday physical systems exhibit effectively local and statistically independent behavior, despite being ultimately composed of quantum constituents. Any satisfactory account of Bell violations must therefore explain not only their existence, but also their suppression in the classical limit.

Within the present framework, this transition is understood as a change in the effective properties of the projection from underlying configurations to observable descriptions. While the projection is generally non-injective at microscopic scales, environmental interactions, coarse-graining, and limited observational resolution progressively reduce the number of distinct underlying configurations compatible with a given observable outcome. As a result, the effective mapping becomes approximately injective.

Decoherence plays a central role in this process. Through continuous interaction with uncontrolled environmental degrees of freedom, interference between distinct underlying configurations is dynamically suppressed. From the perspective of observable descriptions, this suppression eliminates the global consistency constraints responsible for non-factorizable joint probabilities. Observable outcomes can then be treated as effectively independent, restoring the factorization required for classical statistical reasoning.

In this regime, joint probabilities admit an approximate decomposition of the form

$$P(a, b \mid A, B) \approx P(a \mid A) P(b \mid B), \quad (4)$$

up to corrections that are exponentially small in the strength of environmental coupling or the degree of coarse-graining. Bell inequalities are therefore not violated in practice, not because the underlying structure has changed, but because the effective description no longer retains access to non-injective correlations.

Importantly, this recovery of classical behavior does not require the introduction of additional collapse postulates or modifications of quantum dynamics. It follows from the same structural mechanism responsible for Bell violations, applied in a regime where the projection becomes effectively one-to-one. Classicality thus corresponds to a limit in which observable descriptions admit an approximately complete specification, rendering joint probabilities effectively factorizable.

From this viewpoint, the quantum-to-classical transition reflects a change in descriptive accessibility rather than a change in fundamental physical laws. Bell-type correlations are suppressed when the information loss associated with non-injective projection is masked by decoherence and coarse-graining. The classical world emerges as a regime in which effective injectivity is restored, and with it, the validity of classical notions of separability and locality.

5 Discussion and Conceptual Positioning

The analysis presented in this work proposes a structural reinterpretation of Bell inequality violations that avoids introducing nonlocal dynamics, hidden variables, or modifications of quantum mechanics. It is therefore important to clarify how this approach relates to existing interpretations and foundational proposals addressing quantum nonlocality.

Unlike local hidden-variable models, the present framework does not assume the existence of a set of accessible variables that fully determine observable outcomes. No completion of the quantum state is postulated, and no additional degrees of freedom are introduced. The failure of Bell inequalities arises not from incomplete knowledge of underlying variables, but from the non-injective nature of the mapping between underlying configurations and observable descriptions.

The present approach is also distinct from superdeterministic models. While superdeterminism denies statistical independence between measurement settings and underlying variables, no such assumption is required here. Measurement choices remain free and uncorrelated with the underlying configuration space. Bell violations arise solely from the structural properties of effective descriptions, not from conspiratorial correlations between settings and states.

Retrocausal interpretations similarly invoke influences propagating backward in time to account for nonlocal correlations. By contrast, the mechanism discussed here is entirely atemporal at the descriptive level. No causal influence, forward or backward, is required between distant measurement events. Correlations emerge from the global consistency constraints imposed by non-factorizable effective descriptions, rather than from dynamical signaling.

The present framework is compatible with standard accounts of quantum contextuality, but shifts the emphasis from contextual measurement dependence to descriptive non-injectivity. Rather than attributing Bell violations to context-dependent properties of observables, we identify a more general structural origin: the impossibility of assigning independent local outcomes within a single effective description when the underlying mapping is many-to-one.

Finally, we stress that the present work does not advocate a specific ontological interpretation of quantum mechanics. It neither endorses nor rejects realism, operationalism, or informational approaches. Our aim is more modest: to isolate a minimal and sufficient structural condition under which Bell-type inequalities fail. Non-injective projection provides such a condition, clarifying how quantum correlations can arise without abandoning locality at the underlying level.

In this sense, Bell inequality violations may be viewed not as evidence for exotic dynamics, but as indicators of intrinsic limitations of effective descriptions. They reveal the boundaries of applicability of classical probabilistic reasoning, rather than the presence of nonlocal causal mechanisms.

6 Discussion and Conceptual Positioning

6.1 Relation to local hidden-variable models

Violations of Bell inequalities are commonly interpreted as ruling out local hidden-variable models. More precisely, Bell's theorem constrains a class of effective descriptions in which measurement outcomes are assumed to be determined by a set of underlying variables that renders joint probabilities factorizable for spacelike separated measurements.

The analysis presented here clarifies that Bell inequalities do not directly constrain underlying physical dynamics. They constrain the logical structure of effective probabilistic descriptions. When observable outcomes admit an injective or effectively factorizable representation in terms of underlying variables, Bell-type inequalities apply. When this condition fails, Bell-type correlations may arise even if the underlying description is local, deterministic, and causally well behaved.

Within the framework considered here, the failure of Bell inequalities originates from the non-injective nature of the mapping between underlying configurations and observable outcomes. Because multiple underlying configurations correspond to the same effective description, no set of outcome-determining variables exists at the observable level that would allow joint probabilities to be decomposed into independent local contributions. Bell inequalities are therefore violated not because of nonlocal dynamics, but because the assumptions required for probabilistic factorization are not satisfied by the effective description itself.

6.2 Non-injective projection versus superdeterminism and free choice

A frequent objection to any violation of statistical independence in Bell-type analyses is that it implicitly relies on superdeterministic assumptions, namely correlations between

measurement settings and underlying physical states. In such scenarios, the apparent violation of Bell inequalities is attributed to a conspiratorial coordination between experimental choices and microscopic conditions.

The mechanism discussed here is fundamentally different. No correlation is postulated between measurement settings and underlying configurations, and no restriction is imposed on the freedom of experimental choices at the level of observable descriptions. Measurement settings are assumed to be freely and independently chosen within the effective description, in full accordance with standard experimental practice.

The loss of statistical independence arises instead from the non-injective structure of the mapping from underlying configurations to observable outcomes. Because distinct underlying configurations are identified at the effective level, conditioning on observable variables does not isolate independent subsystems. The resulting failure of probabilistic factorization is therefore structural and geometric in origin, rather than causal or conspiratorial.

In this sense, the present framework preserves operational free choice while violating the assumptions required for Bell-type factorizations. The breakdown of statistical independence reflects intrinsic limitations of effective descriptions, not a fine-tuned coordination of initial conditions.

6.3 Measurement, contextuality, and effective descriptions

The structural origin of Bell-type correlations identified above naturally raises the question of their suppression in classical macroscopic regimes. Everyday physical systems exhibit effectively local and statistically independent behavior, despite being composed of quantum constituents.

Within the present framework, this transition is understood as a change in the effective properties of the descriptive mapping. Environmental interactions, decoherence, and coarse-graining progressively reduce the number of distinct underlying configurations compatible with a given observable outcome. As a result, the effective mapping becomes approximately injective, and joint probabilities recover an approximately factorizable form.

In this regime, observable outcomes may be treated as effectively independent, and Bell inequalities are no longer violated in practice. Classical statistical independence thus emerges as a limiting case of effective injectivity, rather than as a fundamental principle.

The present work does not propose a modification of quantum mechanics or a new ontological framework. Its contribution is to isolate a minimal and sufficient structural condition for the failure of Bell inequalities, thereby clarifying the conceptual origin of quantum correlations while preserving both operational locality and empirical adequacy.

We conclude that Bell inequality violations need not be interpreted as evidence for fundamental nonlocality, but may instead reflect intrinsic limitations of effective descriptions in which observable outcomes are treated as independently specifiable local properties.

Appendices

A Formal Conditions for the Failure of Probabilistic Factorization

This appendix provides a formal clarification of the structural conditions under which probabilistic factorization fails in the presence of non-injective projections. The purpose is not to introduce new physical assumptions, but to make explicit the logical content underlying the arguments presented in the main text.

Let (Ω, Σ, μ) be a probability space describing underlying configurations $\omega \in \Omega$, equipped with a normalized measure μ . Observable outcomes are obtained through a measurable mapping

$$\Pi : \Omega \rightarrow \mathcal{O}, \quad (5)$$

where $\mathcal{O}$ denotes the space of observable descriptions. We assume that Π is non-injective, so that there exist distinct $\omega_1 \neq \omega_2$ such that $\Pi(\omega_1) = \Pi(\omega_2)$.

Let A and B denote two spacelike separated measurement settings, and let $a \in \mathcal{O}_A$ and $b \in \mathcal{O}_B$ denote the corresponding observable outcomes. The joint probability distribution accessible at the effective level is given by

$$P(a, b \mid A, B) = \mu(\Pi^{-1}(a, b)) = \sum_{\omega \in \Pi^{-1}(a, b)} \mu(\omega). \quad (6)$$

Suppose that a factorizable representation of the joint distribution exists. Then there must exist a measurable variable λ and a probability density $\rho(\lambda)$ such that

$$P(a, b \mid A, B) = \int d\lambda \rho(\lambda) P(a \mid A, \lambda) P(b \mid B, \lambda). \quad (7)$$

This representation implicitly assumes that λ provides a complete and non-redundant parameterization of the relevant underlying configurations for the observable outcomes.

However, when Π is non-injective, no such variable λ can generically exist at the effective level. Any candidate λ constructed from observable data must identify entire equivalence classes

$$[\omega] = \{\omega' \in \Omega \mid \Pi(\omega') = \Pi(\omega)\}, \quad (8)$$

rather than individual underlying configurations. As a result, conditioning on λ does not isolate independent contributions associated with A and B .

More formally, if λ is a function of $\Pi(\omega)$, then λ is constant over each equivalence class $[\omega]$. The joint probability conditioned on λ therefore takes the form

$$P(a, b \mid A, B, \lambda) = \sum_{\omega \in [\omega_\lambda]} \mu(\omega \mid \lambda), \quad (9)$$

which does not decompose into a product of independent marginal terms unless additional constraints are imposed. In generic non-injective cases, such constraints are absent, and factorization fails.

This establishes that non-injectivity of the mapping from underlying configurations to observable descriptions is sufficient to obstruct any representation of the form Eq. (7). The failure of Bell-type factorization thus follows from a purely structural property of the descriptive mapping, independently of any assumptions about dynamical nonlocality, retrocausality, or hidden variables.

The result emphasizes that Bell inequalities apply only to effective descriptions admitting an injective or equivalently factorizable representation. When observable outcomes arise through non-injective projections, violations of Bell inequalities are a natural and generic consequence of the descriptive structure itself.

B A Toy Model Illustrating Non-Injective Projection

We present a simple toy model intended to illustrate how Bell-type correlations may arise from non-injective projection without invoking nonlocal dynamics, hidden variables, or superdeterministic assumptions. The model is deliberately minimal and does not aim to reproduce the full structure of quantum theory. Its purpose is solely to make explicit the logical mechanism underlying the failure of probabilistic factorization.

B.1 Underlying configuration space

Let Ω denote a finite set of underlying configurations. Each configuration $\omega \in \Omega$ represents a complete specification of a relational structure. No spatiotemporal interpretation is assumed at this level, and no notion of subsystem decomposition is imposed.

We consider a situation in which two measurement regions, denoted A and B , are identified only at the level of observable descriptions. At the level of Ω , the configurations need not decompose into independent components associated with A and B .

B.2 Non-injective projection

Observable outcomes are defined through a projection

$$\Pi : \Omega \rightarrow O_A \times O_B,$$

where O_A and O_B denote the sets of possible outcomes for measurements performed in regions A and B , respectively.

The projection Π is assumed to be non-injective. Distinct underlying configurations may therefore correspond to the same pair of observable outcomes (a, b) . Observable states are thus identified with equivalence classes

$$[\omega] = \{\omega' \in \Omega \mid \Pi(\omega') = \Pi(\omega)\}.$$

B.3 Emergence of correlations

Let $\mu(\omega)$ be a probability distribution defined on Ω . The joint probability distribution accessible at the observable level is given by

$$P(a, b) = \sum_{\omega \in \Pi^{-1}(a, b)} \mu(\omega).$$

Because the preimage $\Pi^{-1}(a, b)$ does not generically decompose into independent subsets associated with A and B , the joint probability $P(a, b)$ does not admit a factorized representation of the form $P(a)P(b)$ or $P(a | \lambda)P(b | \lambda)$ for any effective variable λ .

The resulting correlations arise not from any interaction or information exchange between A and B , but from the fact that both outcomes are constrained by the same underlying relational configuration. From the perspective of the effective description, the two outcomes appear correlated even when the corresponding measurements are spacelike separated.

B.4 Free choice and absence of superdeterminism

Measurement settings at A and B may be freely and independently chosen within the effective description. The probability distribution $\mu(\omega)$ is assumed to be independent of these choices. No correlation between measurement settings and underlying configurations is required or assumed.

The violation of statistical independence arises solely from the non-injective nature of the projection Π . Because observable outcomes do not provide access to the full underlying configuration, conditioning on observable variables does not isolate independent degrees of freedom. The failure of probabilistic factorization is therefore structural rather than conspiratorial.

B.5 Interpretation

This toy model illustrates how two observables may appear correlated because they represent different projections of a single underlying structure. In this sense, Bell-type correlations may be understood as consistency constraints imposed by a non-factorizable effective description, rather than as signatures of nonlocal causation.

While highly simplified, the model captures the essential logical mechanism discussed in the main text. More elaborate realizations may be constructed by enriching the structure of Ω or the projection Π , but the origin of the correlations remains the same: the non-injective mapping between underlying configurations and observable outcomes.

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